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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Wireless Regional Area Network (WRAN) Using Cognitive Radio

Presentations by:

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Aim and Objectives:

Main Aim:

To design and simulate a Cognitive Radio-based WRAN system that efficiently detects and utilizes unused spectrum (spectrum holes) using adaptive multi-band energy detection techniques.

Specific Objectives:

To implement multi-band energy detection for identifying spectrum holes

To develop an adaptive CFAR threshold for improved detection accuracy under noise uncertainty

To enable dynamic spectrum access for Secondary Users (SU)

To ensure interference-free communication with Primary Users (PU)

To improve overall spectrum utilization efficiency in WRAN systems

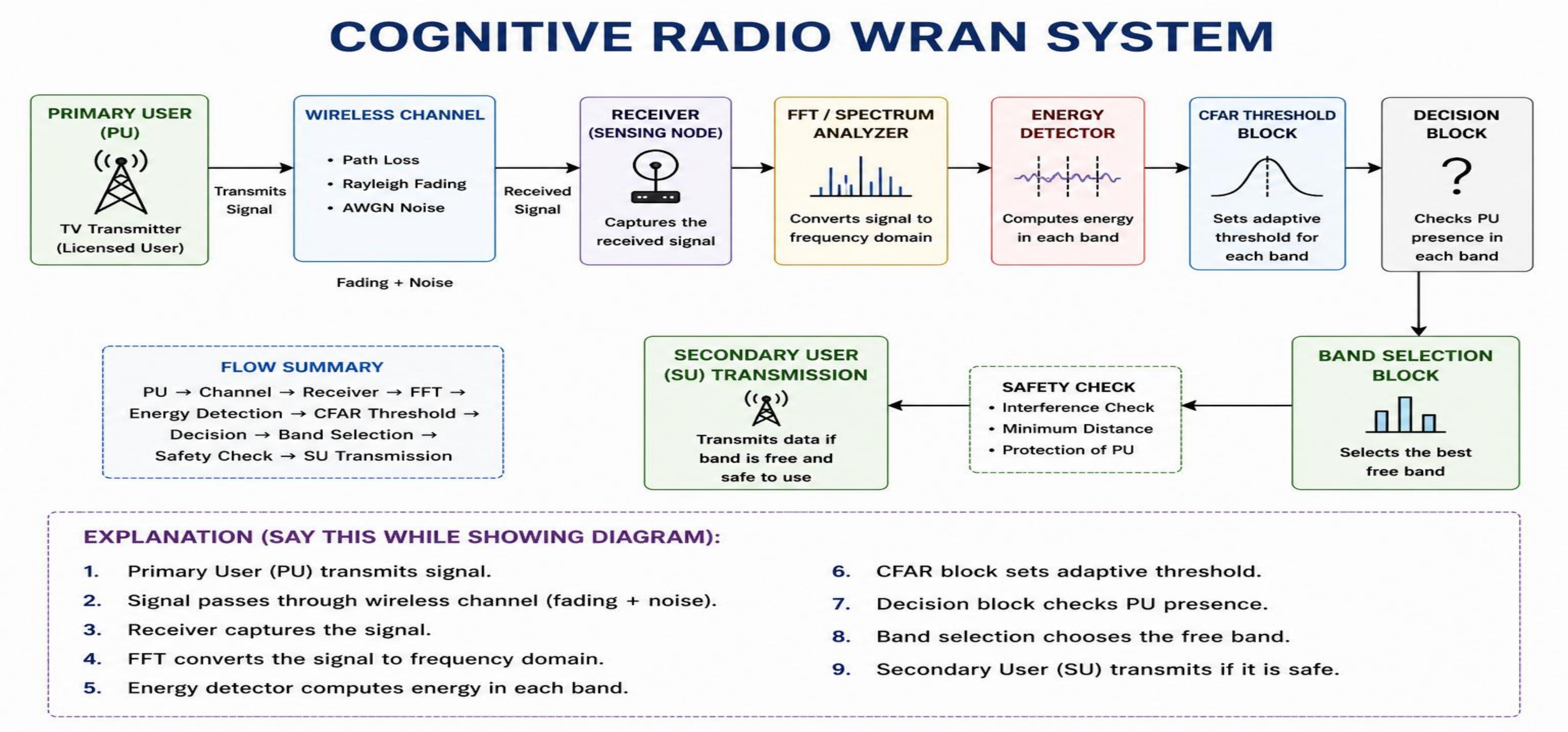
Abstract:

The rapid growth of wireless communication has led to spectrum scarcity, especially in rural areas where traditional technologies face limitations. To address this, the IEEE 802.22 WRAN standard uses Cognitive Radio to utilize underused TV white spaces (54–862 MHz) for broadband communication.

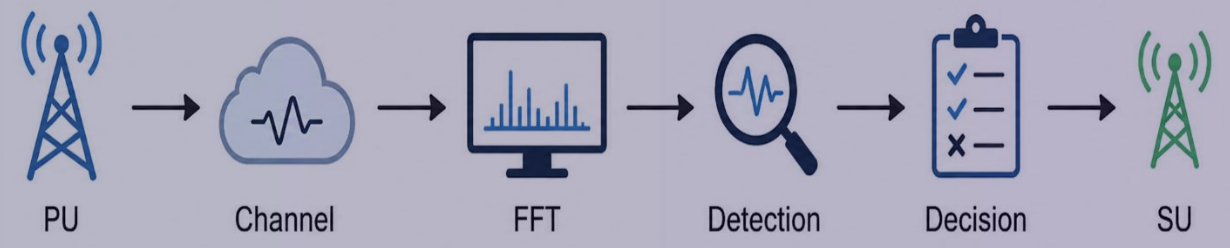
- Implements spectrum hole detection using energy detection
- Uses adaptive CFAR threshold to improve detection under noise uncertainty
- Enables dynamic spectrum access and efficient frequency allocation
- Simulated using MATLAB for performance evaluation
- Supports real-time adaptability and efficient spectrum utilization

PRACTICAL SOLUTION FOR RURAL BROADBAND CONNECTIVITY & EFFICIENT SPECTRUM MANAGEMENT

Block Diagram:



Methodolgy:



- PU & Channel Modeling:

  Markov ON/OFF activity with Rayleigh fading and AWGN noise

- Spectrum Analysis & Detection:

  FFT-based multi-band processing with energy detection on each band

- Adaptive Thresholding & Decision:

  CFAR threshold adapts to noise and identifies free bands

  Interference Control & Transmission:

  Distance check ensures no PU interference and allows safe SU transmission

PU → Channel → FFT → Detection → Decision → SU

Project Setup

- Simulation Environment:

MATLAB used for system design and analysis.

- Signal Processing:

FFT used for multi-band spectrum analysis.

- Detection Technique:

Energy detection with adaptive CFAR threshold.

- Channel Model:

Rayleigh fading with AWGN noise.

- System Model:

Markov model for Primary User (PU) activity.

- Output Analysis:

Results evaluated using ROC curve, energy plots, and band allocation

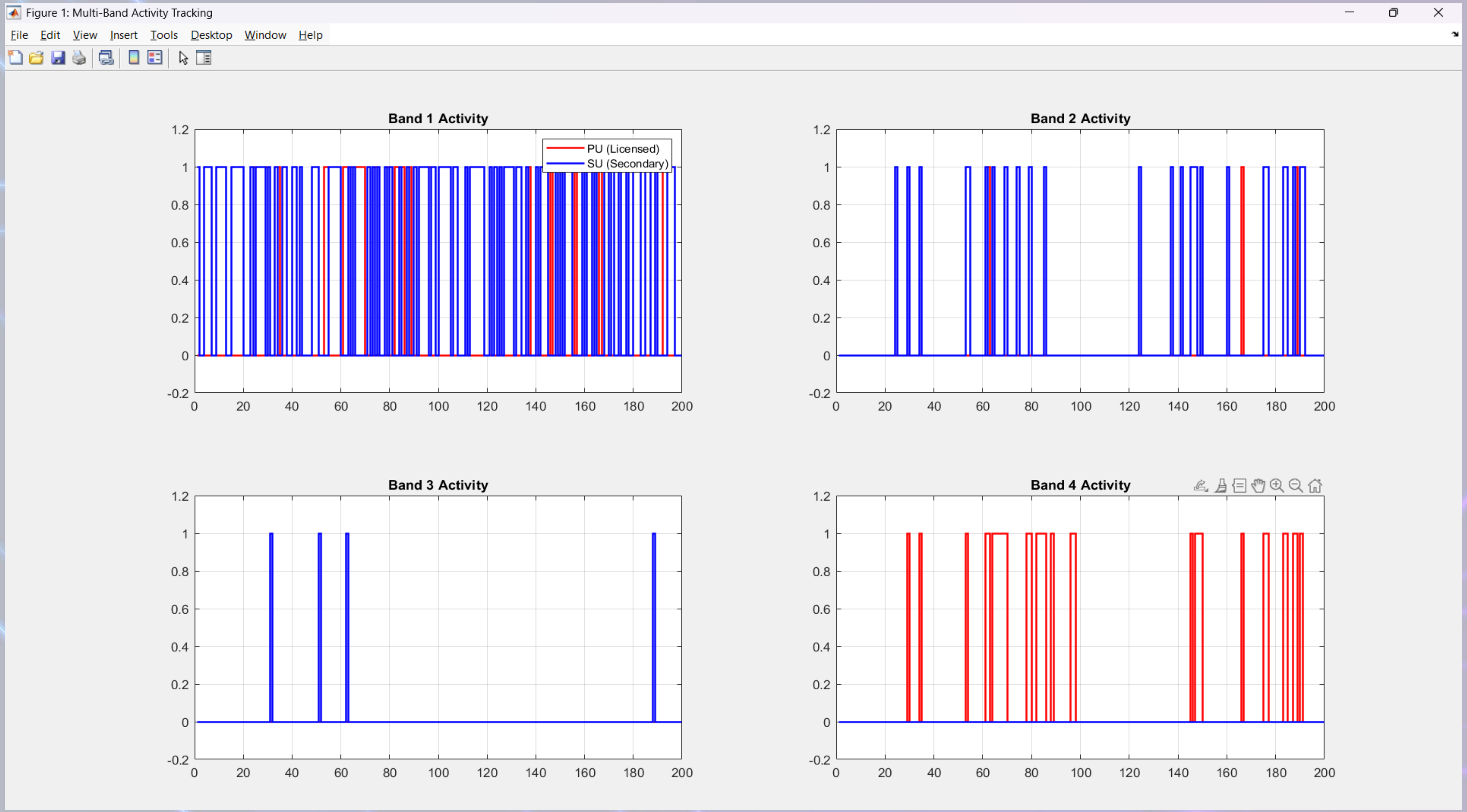
Results

RESULT 1: PU vs SU ACTIVITY

- Legend:
- Red → **PU** Blue → **SU**

Signal Processing:

- SU transmits only when PU is absent
- No overlap → No interference
- Band 1 → High availability
- Band 4 → Mostly occupied



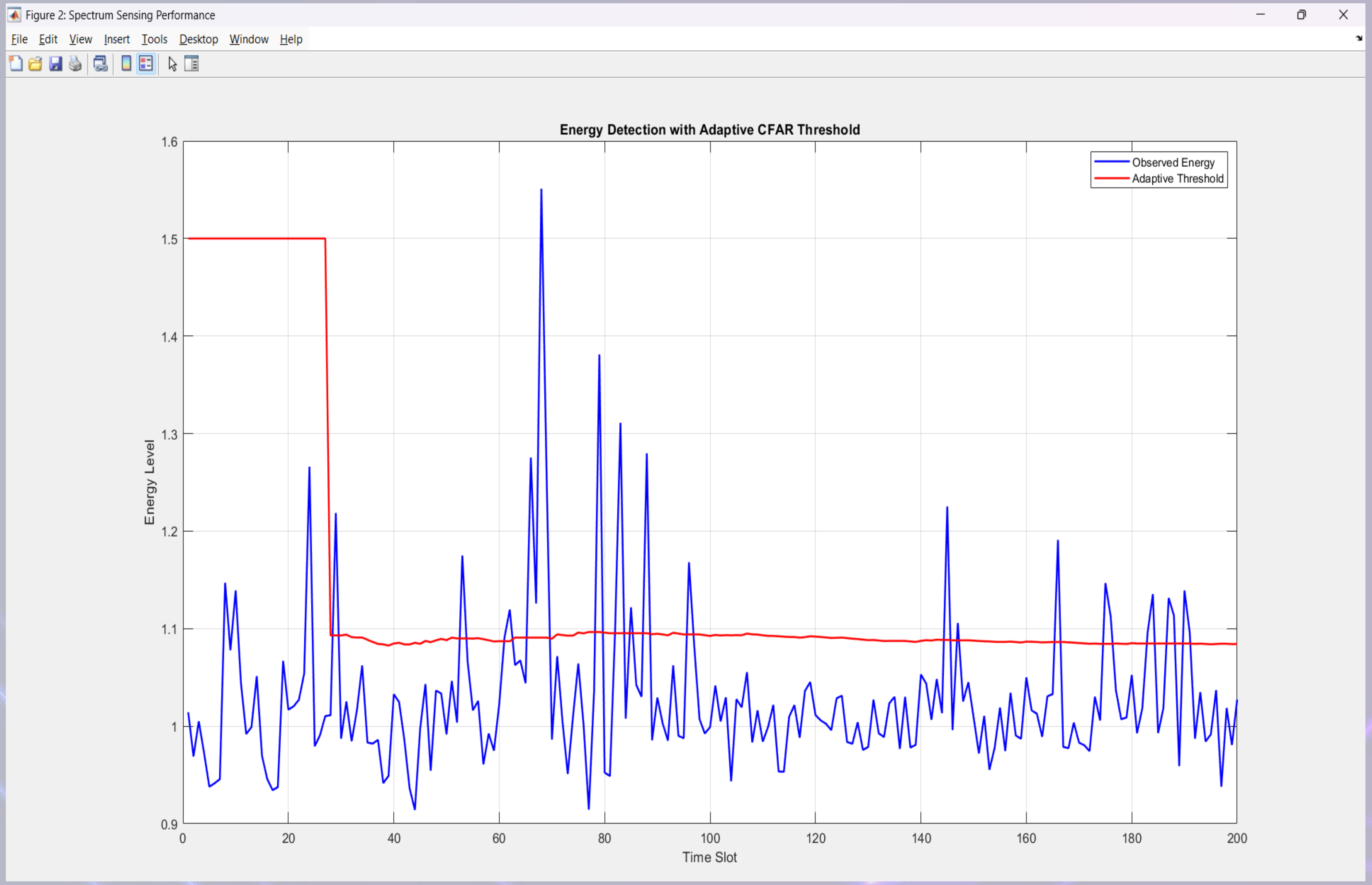
Efficient spectrum utilization

RESULT 2: ENERGY vs THRESHOLD

- Legend:
- Blue → **Signal Energy** Red → **Adaptive Threshold**

Signal Processing:

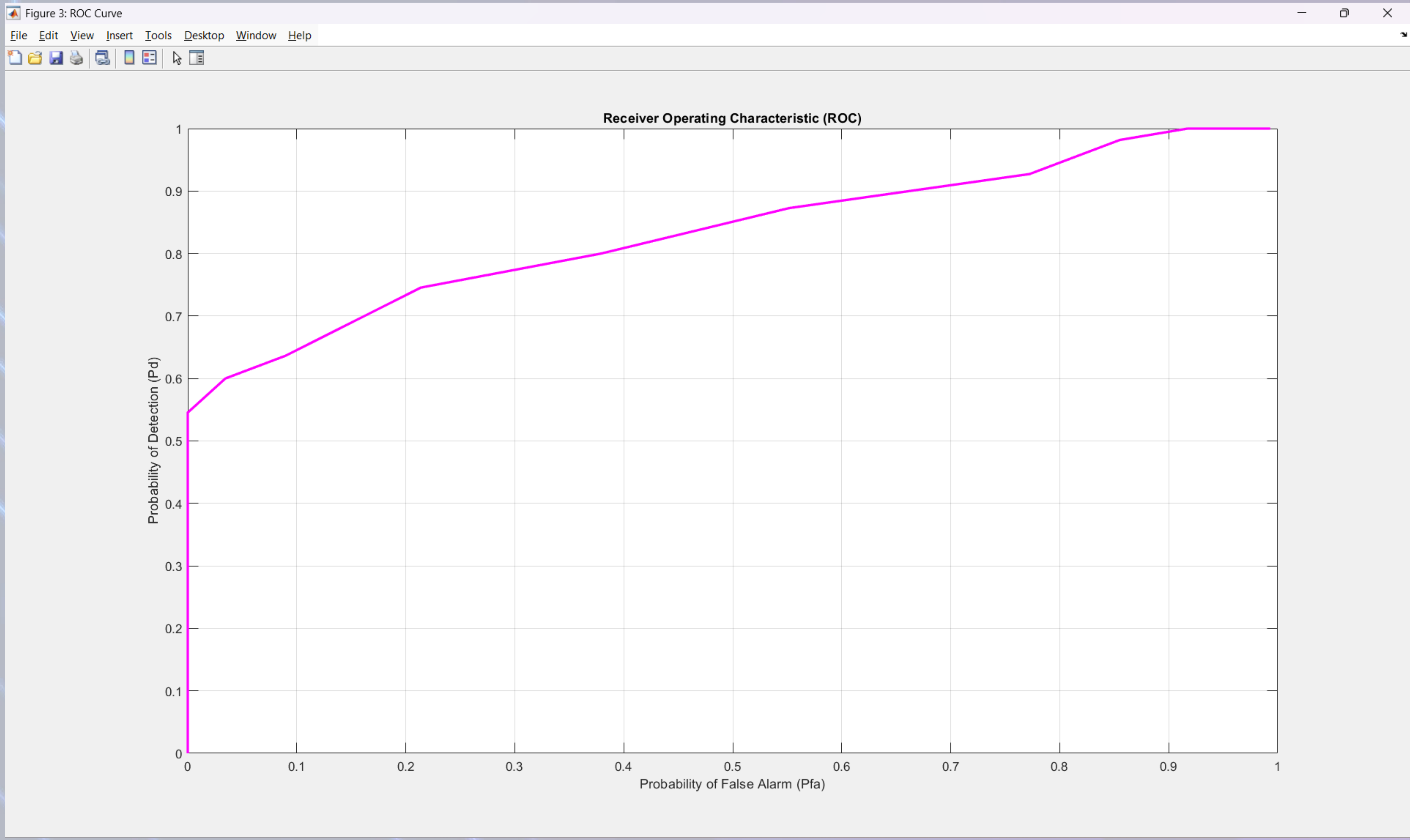
- Threshold adjusts automatically (CFAR)
- Energy > Threshold → PU detected
- Energy < Threshold → Free band



Improved detection accuracy under noise

RESULT 3: ROC CURVE

- Axes:
- X-axis → False Alarm Probability (Pfa)
- Y-axis → Detection Probability (Pd)

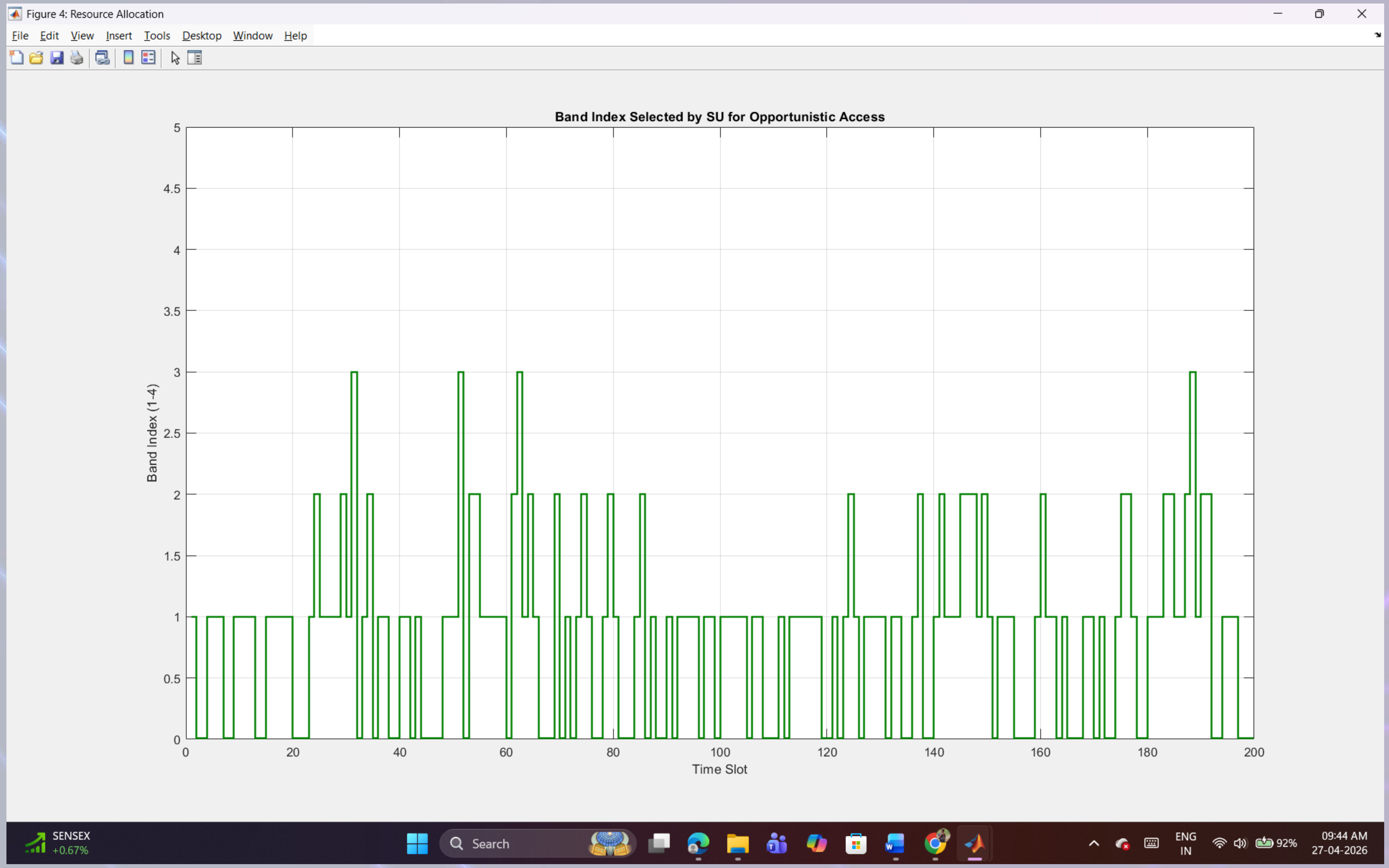


- High Pd even at low Pfa
- Good performance at low SNR (-10 dB)

Strong detection performance

RESULT 4: BAND SELECTION

- SU dynamically selects available band
- Switches band when PU becomes active
- Ensures continuous communication



Intelligent spectrum allocation

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[5] MATLAB Documentation, Signal Processing Toolbox.

Thank you!